

```

//=====
// REG/WIRE declarations
//=====

senhaPac_t  DIGITOS_VALUE;

wire CLK_1K;


//=====
// Structural coding
//=====

divfreq div(
    .reset(!KEY[0]),
    .clock(CLOCK_50),
    .clk_i(CLK_1K)
);


segment7 CONV_0(.bcd(DIGITOS_VALUE[3:0]), .seg(HEX0));
segment7 CONV_1(.bcd(DIGITOS_VALUE[7:4]), .seg(HEX1));
segment7 CONV_2(.bcd(DIGITOS_VALUE[11:8]), .seg(HEX2));
segment7 CONV_3(.bcd(DIGITOS_VALUE[15:12]), .seg(HEX3));
segment7 CONV_4(.bcd(DIGITOS_VALUE[19:16]), .seg(HEX4));
segment7 CONV_5(.bcd(DIGITOS_VALUE[23:20]), .seg(HEX5));


decodificador_de_teclado my_teclado (
    .clk(CLK_1K),
    .rst(!KEY[1]),
    .enable(KEY[2]),
    .col_matriz({GPIO_0[16],GPIO_0[14],GPIO_0[12],GPIO_0[10]}),
    .lin_matriz({GPIO_0[24],GPIO_0[22],GPIO_0[20],GPIO_0[18]}),
    .digitos_value(DIGITOS_VALUE),
    .digitos_valid(LED_R[9])
);

```

```
module segment7( bcd, seg );

    input [3:0] bcd;
    output [6:0] seg;
    reg [6:0] seg;

    always @(bcd)
    begin
        case (bcd)
            0 : seg = 7'b1000000;
            1 : seg = 7'b1111001;
            2 : seg = 7'b0100100;
            3 : seg = 7'b0110000;
            4 : seg = 7'b0011001;
            5 : seg = 7'b0010010;
            6 : seg = 7'b0000010;
            7 : seg = 7'b1111000;
            8 : seg = 7'b0000000;
            9 : seg = 7'b0011000;
            10 : seg = 7'b0001000;
            11 : seg = 7'b0000011;
            12 : seg = 7'b1000110;
            13 : seg = 7'b0100001;
            14 : seg = 7'b0000110;
            15 : seg = 7'b0001110;
            default : seg = 7'b1111111;
        endcase
    end
endmodule
```

```
decodificador_de_teclado my_teclado(
    .clk(),
    .rst(),
    .enable(),
    .col_matriz(),
    .lin_matriz(),
    .digitos_value(),
    .digitos_valid()
);

endmodule
```

```
module divfreq(input reset, clock, output logic clk_i);
    int cont;
    always @(posedge clock or posedge reset) begin
        if(reset) begin
            cont = 0;
        end
    end
endmodule
```

```
    clk_i = 0;
end
else
    if( cont <= 25000 )
        cont++;
    else begin
        clk_i = ~clk_i;
        cont = 0;
    end
end
end
endmodule
```